Causal dimension reduction for environmental health



Wake Forest University BDS Faculty Candidate Talk

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Environmental mixtures analysis

How can we quantify the causal effect of complex environmental exposures?

- Air pollution mixtures ($PM_{2.5}$, NO_2, O_3, ...)
- PFAS compounds (PFOA, PFOS, PFNA, ...)
- Metals (Pb, Cd, As, Hg, ...)

All involve multiple, highly correlated, continuous exposures



EDITORIAL • Environ Health Perspect. 2013 Jan 1;121(1):a6–a8. doi: 10.1289/ehp.1206182

Unraveling the Health Effects of Environmental Mixtures: An NIEHS Priority

Danielle J Carlin [1], Cynthia V Rider [1], Rick Woychik [1], Linda S Birnbaum [1]

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Traditionally, toxicological studies and human health risk assessments have focused primarily on single chemicals. However, people are exposed to a myriad of chemical and nonchemical stressors every day and throughout their lifetime. Some recent events that highlight the need to understand these complex exposures and their role in the etiology of disease include the *Deepwater Horizon* oil spill, the earthquake in Japan and subsequent Fukushima nuclear disaster, and the unknown environmental and health effects of hydraulic

Standard setup of the observed data

Observed data: n-vectors $O_1, ..., O_n$ sampled i.i.d. where

$$O = (Y, X, C) \sim P \in \mathcal{M}$$

- $Y \in \mathbb{R}$: outcome (binary, count, or continuous)
- $X \in \mathbb{R}^p$: continuous exposure mixture
- $C \in \mathbb{R}^q$: measured baseline covariates

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Question: What is the value of Y had we intervened and set $X = x$?

How are we currently answering this question?

Let Y^x be the potential outcome of Y after setting $X = x$.

We are interested in estimating $\mathbb{E}(Y^x)$, the **causal average dose-response**.

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Problem: Each has their pros and cons, but none of these are estimating $\mathbb{E}(Y^x)$.

Causal inference for continuous exposures

Flexible, nonparametric, doubly robust estimators of $\mu(x) := \mathbb{E}(Y^x)$ are available.

Let $m(x, c) := \mathbb{E}(Y \mid X = x, C = c)$ and $\pi(x \mid c) := f_{X|C}(x \mid c)$.

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A kernel-based doubly robust estimator is

$$\hat{\mu}_{DR}(x) = \frac{1}{n} \sum_{i=1}^{n} \left\{ \frac{K_h(X_i - x)}{\hat{\pi}(X_i \mid C_i)} [Y_i - \widehat{m}(X_i, C_i)] + \widehat{m}(x, C_i) \right\},$$

where $K_h(u) = h^{-p} K(u/h)$, and $\widehat{m}$ and $\hat{\pi}$ are estimated flexibly using machine learning (e.g., neural network, random forest, or ensembles).

What can go wrong with multiple exposures?

The doubly robust estimator only works well if we do a good job on $m(x, c)$ and $\pi(x \mid c)$.

$$\sqrt{nh^p} \cdot \|\widehat{m} - m\| \cdot \|\widehat{\pi} - \pi\| \xrightarrow{p} 0.$$

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Solution: Consider estimating the average dose-response in a lower dimensional space.

$$\mathbb{E}(Y^x) = g(\beta^\top x), \qquad \beta \in \mathbb{R}^{p \times d}.$$

Sufficient dimension reduction (SDR)

There exists a rich literature on estimating β that preserves the *regression* relationship:

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Since β is not uniquely identified, the statistical estimand is the **central mean subspace**, $\mathcal{S}_{\mathbb{E}(Y|X)}$, the smallest subspace satisfying this identity.

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Once $\mathcal{S}_{\mathbb{E}(Y|X)}$ is estimated, we can then

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Problem: Because $\mathbb{E}(Y^x) \neq \mathbb{E}(Y \mid X = x)$ due to confounding, we cannot directly use regression SDR for our causal dose-response.

Ideal workflow of causal sufficient dimension reduction

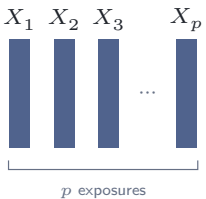
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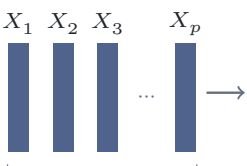


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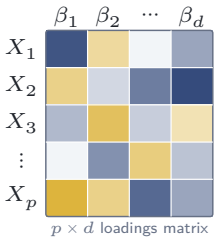
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2. Causal directions

$\beta \in \mathbb{R}^{p \times d}$

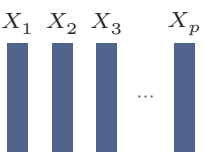


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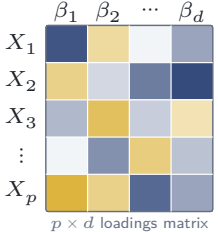
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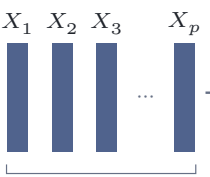
$d \ll p$ summaries

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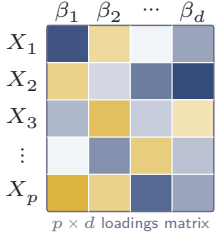
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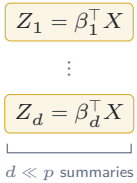
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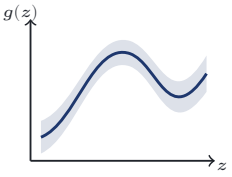
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$\mathbb{E}(Y^x) = g(z)$

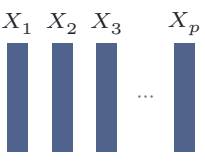


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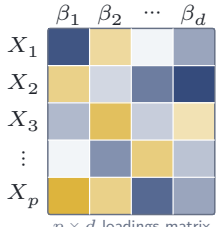
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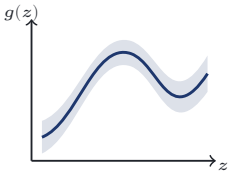
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Question: How can we estimate $\mathcal{S}_{\mathbb{E}(Y^x)}$ when $\mathbb{E}(Y^x) = g(\beta^\top x)$?

Estimation

Marginal structural model approach

Using the theory of marginal structural models (Robins, 2000), Nabi et al. (2022) proposed estimating $\mathcal{S}_{\mathbb{E}(Y^x)}$ by solving the following efficient score:

$$
\begin{aligned}
0 = \frac{1}{n}\sum_{i=1}^{n}\Bigg[& \frac{\pi^*(X_i)}{\pi(X_i \mid C_i)}\Big\{Y_i - \mathbb{E}(Y \mid X_i, C_i)\Big\} \\
& \times \Big\{\alpha(X_i) - \mathbb{E}_q[\alpha(X) \mid \beta^\top X = \beta^\top X_i]\Big\} \\
& + \int_{\mathcal{X}} \Big\{\mathbb{E}(Y \mid X = x, C_i) - \mathbb{E}_q[Y \mid \beta^\top X = \beta^\top x]\Big\} \\
& \times \Big\{\alpha(x) - \mathbb{E}_q[\alpha(X) \mid \beta^\top X = \beta^\top x]\Big\}\pi^*(x)\,dx\Bigg].
\end{aligned}
$$

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While the method is elegant, practical implementation is difficult.

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Nuisance functions to be estimated:

1. $m(x, c) := \mathbb{E}(Y \mid X, C)$
2. $\pi(x \mid c) := f_{X|C}(x \mid c)$
3. $\mathbb{E}_q(Y \mid \beta^\top X = z)$
4. $\mathbb{E}_q[\alpha(X) \mid \beta^\top X = z]$

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Problem: Optimization is challenging when the parameter is matrix-valued and nuisance functions must be re-estimated at each iteration of β.

A simpler approach: recast the dose-response as a regression

Suppose we can construct a variable $\widetilde{Y}$ such that

$$\mathbb{E}(\widetilde{Y} \mid X = x) = \mathbb{E}(Y^x), \qquad \forall x \in \mathcal{X}.$$

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Then under our central assumption $\mathbb{E}(Y^x) = g(\beta^\top x)$, it follows that

$$\mathcal{S}_{\mathbb{E}(\widetilde{Y}|X)} = \mathcal{S}_{\mathbb{E}(Y^x)}.$$

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Solution: Just run standard SDR for regression on $(\widetilde{Y}, X)$ to estimate the causal $\mathcal{S}_{\mathbb{E}(Y^x)}$!

Estimation of $\widetilde{Y}$

Two possible approaches:

1. Causal dose-response:
$$\widetilde{Y} = \mu(X).$$
2. Doubly robust pseudo-outcome:
$$\widetilde{Y} = \xi = \frac{\pi(X)}{\pi(X \mid C)}\{Y - m(X, C)\} + \mathbb{E}_C[m(X, C)].$$

Both require estimation of the same nuisances $m(x, c) = \mathbb{E}(Y \mid X = x, C = c)$ and $\pi(x \mid c) = f_{X|C}(x \mid c)$.

Estimation of $\mathcal{S}_{\mathbb{E}(Y|X)}$ with MAVE

The minimum average variance estimator (MAVE) from Xia et al. (2002) estimates β and $\mathcal{S}_{\mathbb{E}(Y|X)}$ through a variation of local linear regression.

$$\min_{\beta^\top\beta=I_d,\,\{a_j,b_j\}} \sum_{j=1}^{n}\sum_{i=1}^{n}\Big(Y_i - \{a_j + b_j^\top\beta^\top(X_i - X_j)\}\Big)^2 w_{ij}$$

where $w_{ij} = K_h(X_i - X_j)/\sum_{\ell=1}^{n} K_h(X_i - X_\ell)$.

Our CSDR Approach

CSDR Algorithm:

1. Use $O = (Y, X, C)$ to estimate $\widetilde{Y}$
2. Estimate $\mathcal{S}_{\mathbb{E}(Y^x)}$ by running MAVE on $(\widetilde{Y}, X)$
3. Estimate $\mu(x)$ by replacing X with $Z := \widehat{\beta}^\top X$

$$\hat{\mu}_{DR,\widehat{\beta}}(z) = \frac{1}{n}\sum_{i=1}^{n}\left\{\frac{K_h(\widehat{Z}_i - z)}{\hat{\pi}_{\widehat{\beta}}(\widehat{Z}_i \mid C_i)}\left[Y_i - \widehat{m}_{\widehat{\beta}}(\widehat{Z}_i, C_i)\right] + \widehat{m}_{\widehat{\beta}}(z, C_i)\right\},$$

Alternative approach

The CSDR approach improves on the MSM approach by avoiding numerical optimization on β-dependent nuisances.

However, estimation of $\widetilde{Y}$ still involves nuisance estimation of $m(x, c)$ and $\pi(x \mid c)$.

We propose another alternative that can estimate $\mathcal{S}_{\mathbb{E}(Y^x)}$ without $m(x, c)$ and $\pi(x \mid c)$.

Partial dimension reduction on $\mathbb{E}(Y \mid X, C)$

For this approach, we make an additional assumption that

$$\mathbb{E}(Y^x \mid C = c) = g(\beta^\top x, c).$$

One can show that any β satisfying the above also satisfies our original central assumption:

$$\mathbb{E}(Y^x) = g(\beta^\top x).$$

It follows that the partial central mean subspace of Y given (X, C) is equal to $\mathcal{S}_{\mathbb{E}(Y^x)}$.

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Solution: Run partial dimension reduction directly on the regression $\mathbb{E}(Y \mid X, C)$.

Theoretical results

Let $\delta_n = \sqrt{\log n/(nh^p)}$ and $R_n = \|\hat{\mu}(X) - \mu(X)\|_\infty$ (the error in estimating $\widetilde{Y}$).

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1. **Subspace estimation**: We establish convergence rates for estimating $\mathcal{S}_{\mathbb{E}(Y^x)}$:

- Method 1: $\|\hat{\beta} - \beta\| = O_p(h^3 + h\delta_n + h^{-1}R_n)$
- Method 2: $\|\hat{\beta} - \beta\| = O_p(h^3 + h\delta_n + h^{-1}\delta_n^2)$

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2. **Structural dimension selection**: A cross-validation procedure based on MAVE consistently estimates the dimension of $\mathcal{S}_{\mathbb{E}(Y^x)}$.

$$\mathbb{P}(\hat{d} = d_0) \longrightarrow 1.$$

Theoretical results

One may wonder how estimating $\mu(x) = \mathbb{E}(Y^x)$ using $\hat{\beta}_n$ compares to the true β.

We show that the error from estimating β can be asymptotically negligible:

$$\sqrt{nh_n^d}\{\hat{\mu}_{DR}(x;\hat{\beta}_n) - \hat{\mu}_{DR}(x;\beta)\} = o_P(1)$$

Insight: Estimation of β can achieve $O_p(n^{-1/2})$ convergence, faster than the $O_p((nh^d)^{-1/2})$ rate of $\mu_{DR}(x)$.

Simulations and Real Data

Convergence error and dimension selection

CSDR improves inference of the causal dose-response

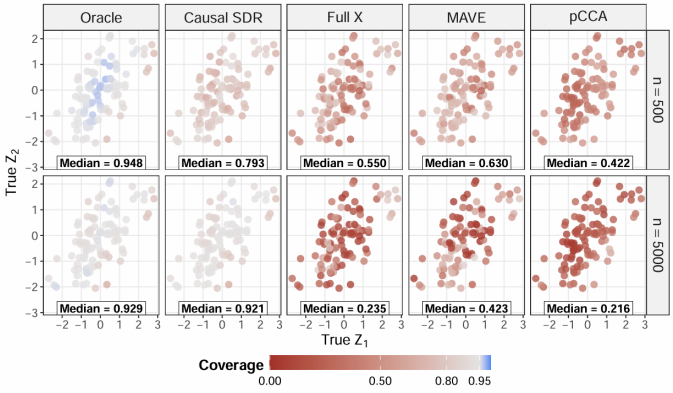


Figure 1: Pointwise 95% coverage for the exposure-response curve ($p = 10$, $d = 2$) at 100 points; methods differ only in β estimation.

PFAS application

We use the **Atlanta African American Maternal-Child Cohort** ($n = 305$) to study the effect of a PFAS mixture ($p = 4$) on infant birthweight. Covariates include maternal age, BMI, education, tobacco use, marijuana use, and infant sex.

ESTIMATED REDUCTION ($\widehat{d} = 1$)

$$\hat{Z} = 0.77\,\text{PFOS} + 0.14\,\text{PFOA} + 0.26\,\text{PFNA} - 0.56\,\text{PFHxS}$$

SUBSPACE IMPORTANCE SCORE $\widehat{\lambda}_j$

PFOS	0.60
PFOA	0.02
PFNA	0.07
PFHxS	0.31

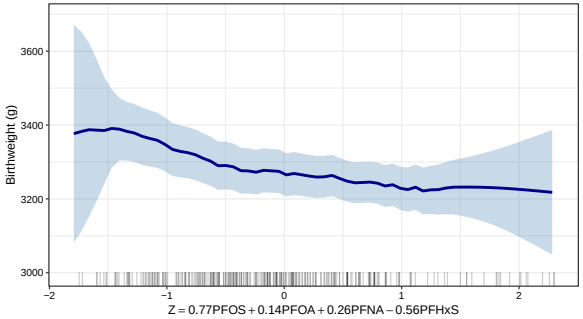
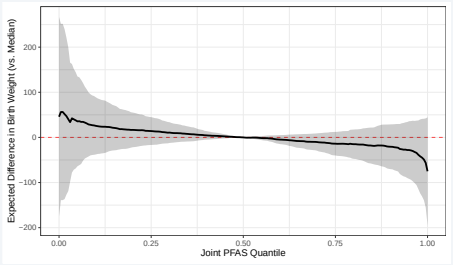


Figure 2: PFAS exposure-response curve from ATL-AA

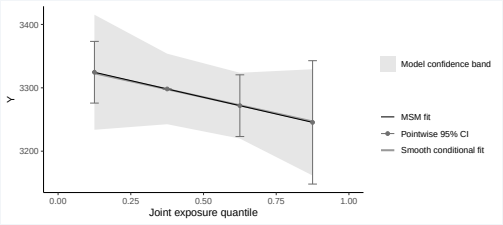
BKMR and qgcomp comparison analysis

BKMR



All four PFAS share a quantile versus the median.

qgcomp



All four PFAS shift together across exposure quartiles.

Takeaway: Different estimands, but in general, increasing PFAS leads to decreasing birthweight.

Future Directions

Causal mediation analysis

Causal representation learning

Causal generative learning

Data Fusion

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Questions?



CSDR preprint

arxiv.org/abs/2606.14840



csdr R package

github.com/tXiao95/csdr

Appendix

Setup outcome, exposures, and confounders

R code

```r
library(csdr)
data(csdr_example2)

Y <- csdr_example2$Y
A <- csdr_example2$A
C <- csdr_example2$C

head(Y, 5)
head(A[, c(1:3, 18:20)], 5)
head(C, 5)
```

Console

```
> head(Y, 5)
     obs1      obs2      obs3      obs4      obs5
  0.3805    0.1838   -2.0486    0.4977    5.4798

> head(A[, c(1:3, 18:20)], 5)
         A1     A2     A3    A18    A19    A20
obs1 -1.066  1.573  0.953  0.033 -0.480  0.500
obs2 -1.028  0.325 -0.364 -0.967 -1.543 -0.572
obs3 -0.800 -0.103 -0.433 -0.011  0.505  0.623
obs4  2.492  0.582 -1.134 -1.473 -1.138  1.551
obs5  1.733 -0.114  0.615  1.083  0.350  0.133

> head(C, 5)
         C1    C2     C3     C4     C5
obs1 -0.935 1.312 -0.294 -1.979  1.150
obs2 -0.694 0.531  0.367 -2.259 -1.293
obs3 -0.548 0.712 -0.088 -0.407  1.828
obs4  0.766 1.050 -0.498 -0.593 -0.604
obs5  1.852 -0.554  0.849  0.643 -0.312
```

Fit the causal subspace

R code

```r
sl_lib <- c(
  "SL.glm", "SL.earth", "SL.ranger"
)

learners <- csdr_learners(
  sl_library = sl_lib
)

set.seed(42)
fit <- csdr(
  Y = Y, A = A, C = C,
  L = 5, seed = 42,
  learners = learners,
  max_dim = 5,
  keep_nuisance = TRUE
)

summary(fit)
```

Console

```
> summary(fit)
Summary of causal sufficient dimension reduction fit

 variant d_hat  n p mave_method target_exposure
      DR     2 100 20   meanMAVE               A

Learners:
 outcome: SuperLearner regression
   family: gaussian
   SL.library: SL.glm, SL.earth, SL.ranger
 gps: MVN GPS with linear conditional mean
   method_gps: linear; density_floor: 1e-16

Target construction:
Effective folds: 5
PO marginalization: crossfit
```

Inspect the fitted subspace and scores

R code

```r
B_hat <- coef(fit)
Z_hat <- scores(fit)

mave <- mave_fits(fit)
dim_selection <- mave$dimension_selection

B_hat[c(1, 2, 20), , drop = FALSE]
sis(fit)
head(Z_hat, 2)
dim_selection
```

Console

```
> B_hat[c(1, 2, 20), , drop = FALSE]
       dir1    dir2
A1  -0.3401  0.5777
A2  -0.3524  0.4195
A20 -0.0521 0.0784

> sis(fit)
    A1    A2    A3    A4    A5    A6    A7    A8    A9   A10
 0.464 0.312 0.533 0.390 0.000 0.004 0.002 0.020 0.001 0.021
   A11   A12   A13   A14   A15   A16   A17   A18   A19   A20
 0.068 0.059 0.028 0.020 0.006 0.005 0.015 0.036 0.006 0.009

> head(Z_hat, 2)
       score1  score2
obs1  -0.1862  0.4515
obs2   1.5224 -0.3484

> dim_selection
Dimensions      0    1    2    3    4    5
CV-value     1.58 1.13 0.75 0.77 0.92 0.97
Selected dimension: 2
```

Estimate the response in the reduced exposure

R code

```r
z_grid <- matrix(
  rep(apply(Z_hat, 2, median), each = 50),
  nrow = 50,
  dimnames = list(NULL, colnames(Z_hat))
)
z_grid[, 1] <- seq(
  quantile(Z_hat[, 1], 0.05),
  quantile(Z_hat[, 1], 0.95),
  length.out = 50
)

ers <- estimate_ers(
  Y = Y, A = Z_hat, C = C,
  a_eval = z_grid, estimator = "DR",
  L = 5, seed = 42,
  verbose = FALSE
)

head(ers$results)
```

Console

```
> head(ers$results)
  score1 score2 estimate     se ci_lower ci_upper effective_n
1 -1.990 0.0015    2.676 0.5599    1.579    3.774       4.502
2 -1.912 0.0015    2.693 0.4762    1.760    3.627       4.559
3 -1.834 0.0015    2.703 0.3920    1.935    3.472       4.671
4 -1.756 0.0015    2.703 0.3136    2.088    3.317       4.851
5 -1.678 0.0015    2.687 0.2493    2.198    3.176       5.113
6 -1.601 0.0015    2.652 0.2100    2.241    3.064       5.476
    min_gps h_used_1 h_used_2
1 4.049e-05    0.578    0.596
2 4.049e-05    0.578    0.596
3 4.049e-05    0.578    0.596
4 4.049e-05    0.578    0.596
5 4.049e-05    0.578    0.596
6 4.049e-05    0.578    0.596
```

Dimension Selection

NW leave-one-out estimator

$$\hat{a}_{d0,j} = \frac{\sum_{i \neq j} K_{h_d}^{(i,j)} \tilde{Y}_i}{\sum_{i \neq j} K_{h_d}^{(i,j)}}, K_{h_d}^{(i,j)} = K_{h_d}\{\hat{\beta}_1^\top(\tilde{X}_i - \tilde{X}_j), \ldots, \hat{\beta}_d^\top(\tilde{X}_i - \tilde{X}_j)\}$$

Cross-validated error

$$\mathsf{CV}_n(d) = \frac{1}{n}\sum_{j=1}^{n}\left(\tilde{Y}_j - \hat{a}_{d0,j}\right)^2, \qquad \mathsf{CV}_n(0) = \frac{1}{n}\sum_{i=1}^{n}(\tilde{Y}_i - \bar{Y})^2$$

Estimated structural dimension

$$\hat{d} = \arg\min_{0 \leq d \leq p} \mathsf{CV}_n(d)$$